# Denominator Periods, Rational-Value Constraints and Achievement-Set Geometry

*Mersenne subseries and Erdős Problem #257*

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July 2026


## ABSTRACT

Erdős Problem #257 asks whether $X_A = \sum_{a \in A} (2^a - 1)^{-1}$ is irrational for every infinite $A \subseteq \mathbb{N}_{>0}$. The lead theorem proved here is that for every integer base $b \geq 2$ and every infinite $A \subseteq \mathbb{N}_{>0}$ with $\sum_{a \in A} a^{-1} < \infty$ the value $X_A(b)$ is irrational, with no coprimality hypothesis, and the complete argument is checked by the Lean kernel. Erdős printed the pairwise-coprime case for every integer base $b \geq 2$ in Math. Student 36 (1968), 222–226 and wrote on the same page that the coprimality condition is superfluous, giving no argument; the statement is his, and the proof and its check are the contribution here. The theorem removes the entire convergent-reciprocal-mass regime, so the exact remaining obligation in the universal problem is the reciprocal-divergent supports.

Three further kernel-checked statements describe the hypothetical counterexample rather than sampling supports. A Boolean–Möbius correspondence makes existence of a normalised support of value $p/q$ equivalent to existence of one square-root-bounded integer orbit, and reconstructs the support from that orbit. A uniform repair criterion decides membership in the subsum set for every real target through one finite square-root window, so a single strictly increasing window refutes membership. A scaled greedy recursion identifies the subsum set with the non-escaping set of an explicit nearly doubling map, with exponential escape off it. The classical full-support theorem of Erdős at every integer base is formalised together with its certificate machine, which also supplies arbitrarily long zero blocks in the base-$b$ expansion of that value.

Beyond reciprocal summability we prove irrationality, by ordinary arguments, for several classes that allow divergent reciprocal mass: supports satisfying a divisibility-weighted summability criterion, supports whose reciprocal mass grows more slowly than an iterated logarithm, and supports admitting summable positive fractional divisor covers. Each criterion is hereditary under passage to infinite subsets and works at every integer base $b \geq 2$. A separate argument, using Tao–Teräväinen's correlation theorem, treats every infinite subset of the prime powers, including fixed dilations and finite modifications.

The three extension arguments share a mechanism. Freeze a finite set of exponents by a common multiple, then average the remaining shifted atoms to produce arbitrarily small positive displacements. Rationality forces a fixed positive gap, giving the contradiction. Keeping the incomplete-period error in this average yields a quantitative necessary condition: a rational value forces reciprocal mass to grow at least at an explicit multiple of


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** Authorship and AI use.* Will Cook built and directed the research infrastructure. AI agents produced the mathematical interpretations, proofs, prose, and software. Cook maintains the project and is responsible for its presentation; this is not a claim of personal mathematical verification or independent specialist review.

log-star. Positive fractional covers extend the argument to moving divisor frames, with exponents that may tend to zero.

For finite nonempty supports $F$, the reduced denominator $D_F$ satisfies $\mathrm{ord}_{D_F}(b) = \mathrm{lcm}(F)$; that finite-support theorem supplies no unbounded-support certificate, and it is developed in Section 3. For infinite supports, the achievement set is compact, perfect, nowhere dense, and has Lebesgue measure 1; restricting the exponents gives an exact volume dichotomy. At the rational targets $1/2$ and $1/21$, finite-support exclusion turns membership into a potential counterexample, and exact greedy criteria locate the remaining infinite-orbit question.

The analytic support criteria have ordinary proofs in Section 2 and carry no formal check. The accompanying Lean development checks arbitrary reciprocal-summable support at every integer base, the classical full-support and periodic-weight theorems, the Boolean–Möbius correspondence, the uniform repair criterion, the scaled greedy-trap dynamics, the denominator identity, the geometric classification, and the stated orbit reductions. The unrestricted problem and the two rational membership questions remain open.

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### Main results

**Lead theorem.** Every infinite reciprocal-summable support has irrational value at every integer base, with no coprimality hypothesis; the whole proof is kernel checked. Erdős stated this extension of his printed pairwise-coprime theorem without proof, so the statement is his and the proof is local. What remains of the universal problem is exactly the reciprocal-divergent supports. **Description of every rational value.** A Boolean–Möbius correspondence, a uniform square-root repair criterion valid at every real target, and an exact greedy-trap dynamics for the subsum set are each kernel checked, and each quantifies over every hypothetical counterexample. **Classical theorems formalised.** The full-support theorem of Erdős at every integer base, with zero-block digit witnesses, and the periodic and pairwise-coprime families. **Ordinary extensions.** Divisibility-weighted summability, sub-log-star reciprocal mass, and positive fractional divisor covers each imply irrationality at every integer base, hereditarily. Every infinite prime-power sub-support also has irrational value, by a separate analytic argument. These four have ordinary proofs and no formal check, and they include supports with divergent reciprocal mass. **Finite supports.** The reduced denominator has multiplicative order exactly the least common multiple of the exponents. **Geometry and rational targets.** The subsum set has measure 1 and an exact restricted-support volume law; finite-support exclusion and greedy criteria isolate the unresolved membership questions at $1/2$ and $1/21$.

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## 1 Introduction and main results

**Problem 1.1** (Erdős #257)**.** For every infinite $A \subseteq \mathbb{N}_{>0}$, is $X_A = \sum_{a\in A} \dfrac{1}{2^a - 1}$ irrational?

See Erdős [9, p. 222], Erdős and Graham [10, p. 62], and Erdős [11, p. 105]. Bloom's catalogue records the problem as open [12].

Write $w_n = (2^n - 1)^{-1}$. Expanding each weight as a geometric series and interchanging the two nonnegative sums gives the coordinate this note works in. With the *divisor* *incidence* $\mathrm{sc}_A(n) = \#\{d \mid n : d \in A\}$,

$$X_A = \sum_{a\in A} \frac{1}{2^a - 1} = \sum_{n\geq 1} \frac{\mathrm{sc}_A(n)}{2^n}. \tag{1}$$

The transform is worth reading carefully, because it is where the arithmetic of the problem enters. The datum is the 0/1 indicator of $A$. Equation (1) contains its divisor transform, a nonnegative integer sequence bounded by the divisor function, $\mathrm{sc}_A(n) \leq d(n)$. So #257 is not a generic question about binary digit sequences. Equation (1) is a power-series representation, not a binary expansion: its coefficients are divisor counts drawn from a single support and may exceed 1. Every theorem below is a statement about sequences of that shape.

A small instance fixes the notation. At $A = \{2, 3\}$ the incidence sequence begins

$$\mathrm{sc}_A(1), \dots, \mathrm{sc}_A(12) = 0, 1, 1, 1, 0, 2, 0, 1, 1, 1, 0, 2,$$

the value 2 occurring at the multiples of 6 and the value 0 at the integers prime to 6, and (1) reads $\frac{1}{3} + \frac{1}{7} = \frac{10}{21}$.

Several statements below hold at every integer base, so we write $X_A(b) = \sum_{a\in A}(b^a - 1)^{-1}$ for the value at an integer base $b \geq 2$, with $X_A = X_A(2)$. Base 2 is the case Problem 1.1 asks about and is meant whenever no base is named.

**Theorem 1.2** (reciprocal-summable supports)**.** *Let $b \geq 2$ be an integer and let $A \subseteq \mathbb{N}_{>0}$ be infinite. If $\sum_{a\in A} a^{-1} < \infty$, then $X_A(b)$ is irrational.*

The complete implication, from the two displayed hypotheses to irrationality, is checked by the Lean kernel; no ordinary step and no finite certificate is left outside that check. The proof first obtains binary LCM–Cesàro close returns for the shifted support atoms. A pointwise displacement bound transfers those returns to every integer radix, where an exact integral tail orbit contradicts rationality.

*Attribution.* On p. 222 of [9] Erdős prints the theorem for pairwise-coprime supports with convergent reciprocal sum, at every integer base $b \geq 2$, and on the same page he says that pairwise coprimality can be removed by a more complicated argument, giving no details; p. 226 repeats that boundary. The statement of Theorem 1.2 is therefore his stated extension. What is local is the proof and its checked development. The argument here is not identified with the one he omitted, and its novelty and priority have not been assessed. The printed pairwise-coprime theorem is retained below as an attributed specialisation.

*Remaining obligation.* Theorem 1.2 settles every infinite support with $\sum_{a\in A} a^{-1} < \infty$. The exact statement still to be proved for Problem 1.1 is irrationality of $X_A$ for every infinite $A$ with $\sum_{a\in A} a^{-1} = \infty$. Sections 2 and 4 work inside that regime, and nothing below exhausts it.

**The classical full support, formalised.** The support $A = \mathbb{N}_{>0}$ carries the oldest theorem in this circle, and the certificate machine that proves it is the engine every support family here rides.

**Theorem 1.3** (full support at every integer base)**.** *Let $b \geq 2$ be an integer. Then $\sum_{n\geq 1}(b^n - 1)^{-1}$ is irrational. Moreover, for every $K \in \mathbb{N}$ there are $N \in \mathbb{N}$ and $z \in \mathbb{Z}$ with*

$$0 < b^N \sum_{n\geq 1} \frac{1}{b^n - 1} - z < b^{-K},$$

*so the base-b expansion of that value contains arbitrarily long blocks of zero digits.*

Irrationality is Erdős's 1948 theorem [8]; the statement and the mathematics are his, and what is local is the formalisation. Both displayed clauses are inside the kernel check. Irrationality is checked, with the base-2 instance for the Erdős–Borwein constant at checked. The engine is a bounded Bertrand and Chinese-remainder first-block frame, a middle-window divisor-pair average with pigeonhole selection, and an explicit parameter closure, supplied as one checked certificate family, and the coefficient abstraction is checked to lose nothing at the coefficient-engine form. The zero-block clause is checked, as the precision-$b^K$ specialisation of the near-integer witnesses checked. Two boundaries. The digit clause concerns zero blocks alone; it gives no normality and no digit-frequency law, and the theorem that the binary block 11 occurs infinitely often in the base-2 value is Campbell's separate result [3, Theorem 1, p. 12; proof pp. 12–24]. And $A = \mathbb{N}_{>0}$ is one support, so this theorem leaves the remaining obligation above untouched.

**A description of every rational value.** The next theorem replaces the search for a rational-valued support by a search for one integer orbit, and it reconstructs the support from that orbit. It is possible because the support can always be read back off the incidence sequence: with $\mu$ the Möbius function, $*$ Dirichlet convolution $(g * h)(n) = \sum_{d|n} g(d)h(n/d)$ and $\mathbf{1}_A$ the indicator of $A$, the identity $\mathrm{sc}_A = \mathbf{1}_A * 1$ inverts to $\mu * \mathrm{sc}_A = \mathbf{1}_A$.

For $p \in \mathbb{Z}$, $q \geq 1$ and $U \colon \mathbb{N} \to \mathbb{Z}$, set

$$Q_U(0) = 0, \qquad Q_U(n) = \frac{2U(n-1) - U(n)}{q} \quad (n \geq 1).$$

Call $U$ an *admissible Boolean–Möbius scaled-tail sequence* for $(p, q)$ if

$$\begin{gathered} U(0) = p, \quad U(N) > 0, \quad U(N) \leq q(2\sqrt{N} + 4) \quad (N \geq 0), \\ q \mid 2U(N) - U(N+1) \quad (N \geq 0), \qquad (\mu * Q_U)(n) \in \{0, 1\} \quad (n \geq 1). \end{gathered} \tag{2}$$

The divisibility condition makes $Q_U$ integral; the last condition says that Möbius inversion of the quotient is the indicator of a set.

**Theorem 1.4** (Boolean–Möbius scaled-tail correspondence)**.** *Let $p \in \mathbb{Z}$ and $q \geq 1$. There exists a support $A$ with $0 \notin A$, with some positive element, and with $X_A = p/q$, if and only if there exists an admissible Boolean–Möbius scaled-tail sequence $U$ for $(p, q)$. In that case the support is recovered as*

$$A = \{n : (\mu * Q_U)(n) = 1\},$$

*with $Q_U$ as in (2).*

The equivalence and the reconstruction are checked, with the two directions at support to orbit and orbit to support. The whole statement is inside the kernel check, with no ordinary step and no finite certificate outside it. Prior art for this correspondence has not been assessed; the register records its novelty as unassessed, and no priority is claimed. The finite example $A = \{2, 3\}$ makes the definition concrete. Here $p/q = 10/21$, and

$$Q_U(1), \ldots, Q_U(6) = 0, 1, 1, 1, 0, 2, \qquad U(0), \ldots, U(6) = 10, 20, 19, 17, 13, 26, 10.$$

Möbius inversion gives $(\mu * Q_U)(1), \ldots, (\mu * Q_U)(6) = 0, 1, 1, 0, 0, 0$, the indicator of $\{2, 3\}$ through that range. This finite calculation is a computation and illustrates the correspondence; the admissible scaled-tail sequence itself is an infinite object, and the finite support is not a counterexample to Problem 1.1.

*The remaining obligation after Theorem 1.4.* The support the theorem produces is not required to be infinite, and a finite support supplies an admissible scaled-tail sequence for its own value, so existence of such a sequence is by itself no counterexample to Problem 1.1. What the equivalence changes is the search space. The obligation it leaves is to decide whether an admissible sequence exists whose reconstructed support is infinite.

**The subsum set and the greedy expansion.** The next two theorems are about the set of values obtainable from the weights $w_n$, so we fix that object here and return to its geometry in Section 8. Write

$$\mathcal{A} = \Big\{ \sum_{n \geq 1} \varepsilon_n w_n : \varepsilon_n \in \{0, 1\} \Big\}$$

for the achievement set, the set of all values of finite and infinite selections alike. For a real $x \geq 0$ the *greedy expansion* runs through $n = 1, 2, \ldots$ carrying a remainder, initially $x$, and at level $n$ subtracts $w_n$ from the remainder when $w_n$ does not exceed it, leaving the remainder unchanged otherwise; a level at which nothing is subtracted is *skipped*, and $A_x$ denotes the set of levels at which a subtraction happens. Write $r_N(x)$ for the remainder after level $N$.

**Theorem 1.5** (uniform repair criterion)**.** *For every real $x \geq 0$, put*

$$P_0(x) = 0, \qquad P_{N+1}(x) = 2P_N(x) + \mathrm{sc}_{A_x}(N+1), \qquad Q_N(x) = \lfloor 2^N x \rfloor - P_N(x),$$

*which the prefix bound makes a nonnegative integer. Then*

$$x \in \mathcal{A} \quad \Longleftrightarrow \quad \forall K\, \exists N \in [K, K + 2\lfloor \sqrt{K} \rfloor + 12) : Q_{N+1}(x) \leq Q_N(x).$$

*Equivalently, repairs occur cofinally; equivalently, $Q_N(x) \leq \mathrm{sc}_{A_x}(N+1)$ cofinally; equivalently, $Q_N(x) \leq N + 1$ cofinally. One strictly increasing window excludes membership.*

*Proof.* If $x \in \mathcal{A}$, the full tail identity and divisor-pair bound give $Q_N(x) \leq 2\sqrt{N} + 4$. Put $s = \lfloor \sqrt{K} \rfloor$ and $T = 2s + 12$. Strict increase at each of the $T$ steps would give $Q_{K+T}(x) \geq T$. But $K + T < (s+4)^2$ gives $Q_{K+T}(x) < 2(s+4) + 4 = T$, a contradiction.

For the converse, the prefix recurrence and the nonnegative next binary floor digit give

$$2Q_N(x) - \mathrm{sc}_{A_x}(N+1) \leq Q_{N+1}(x).$$

A repair therefore forces $Q_N(x) \le \mathrm{sc}_{A_x}(N+1) \le N+1$. If $x \notin \mathcal{A}$, a positive fatal gap $\delta$ persists in the actual greedy remainder; the prefix-tail identity yields $Q_N(x) \ge 2^N\delta - 1$ beyond that fatal rank. This eventually exceeds $N+1$, contradicting a cofinal linear bound. These implications give all the stated equivalences. □

The displayed square-root window equivalence is checked for every real $x \ge 0$, with the three further equivalent forms at cofinal repairs, cofinal load bound and cofinal linear bound, and the exclusion clause at one strict window. The displayed proof above is the ordinary reading of that checked statement, and every clause of the theorem is inside the kernel check. Prior art is unassessed and no priority is claimed.

The finite deadline comes from a growth comparison: strict increases of an integer sequence accumulate linearly, whereas represented tails have only a square-root bound. For $x = 4/9$ this recovers the original repair problem with $Q_N = \lfloor 4 \cdot 2^N/9 \rfloor - P_N$. Exact replay through rank 4096 finds no strict window for that target, which is a computation and not a theorem.

*The remaining obligation after Theorem 1.5.* The theorem identifies the certificate to seek at any given target, and it supplies no occurrence of one. Deciding 1/2 or 1/21 or 4/9 still requires either one strictly increasing window or a cofinal supply of repairs, and neither is proved for any of those targets.

**Theorem 1.6** (scaled-greedy trap, escape, and cofinal return)**.** *For every nonnegative real $x$, writing $y_N(x) = 2^N r_N(x)$ for the scaled greedy remainder,*

$$x \in \mathcal{A} \quad \Longleftrightarrow \quad y_N(x) < 2 \quad \textit{for every } N.$$

*If $x \notin \mathcal{A}$, then $y_N(x) \to +\infty$. Conversely, a single bounded cofinal subsequence already characterizes membership:*

$$x \in \mathcal{A} \quad \Longleftrightarrow \quad \exists B \in \mathbb{R}\ \forall K\ \exists N \ge K: \qquad y_N(x) \le B.$$

With $c_n = 2^n/(2^n - 1)$, the scaled recursion is the explicit nearly doubling map sending $y_N$ to $2y_N - c_{N+1}$ when $c_{N+1} \le 2y_N$ and to $2y_N$ otherwise (checked), so $\mathcal{A}$ is the filled non-escaping set of that map. The trap-set equality is checked, its membership form is checked, the escape theorem is checked, and the bounded-cofinal-return equivalence is checked. Every clause is inside the kernel check. The escape proof locates a fatal rank at which the residual exceeds the complete remaining tail by some $\delta > 0$. That excess persists, so the scaled residual is at least $2^N\delta$ thereafter. This is the hard step in the reverse bounded-return implication: an orbit with even one bounded cofinal subsequence cannot lie outside $\mathcal{A}$. Section 8 records that $\mathcal{A}$ is compact, perfect, totally disconnected, nowhere dense and of Lebesgue measure 1, so the theorem exhibits a fat Cantor set as a non-escaping set. The compactness, perfectness and nowhere density are classical, and Kakeya [1, p. 251] and Kovač and Tao [7, Remark 4.1, p. 13] are the antecedents named in Section 8. Novelty for the dynamical identification itself has not been assessed.

*The remaining obligation after Theorem 1.6.* The theorem gives no orbit. Deciding whether 1/2 or 1/21 lies in $\mathcal{A}$ still requires one bounded cofinal return or one escape, and neither is proved; the rational specialisations are in Section 8.

*Status.* The original problem remains open. Statements marked as Lean-checked refer to the linked propositions accepted by the pinned kernel, with no `sorry`, added axioms, or unchecked evaluation. The note states the remaining mathematical obligations explicitly.

*Companion system context.* The claim and trust boundary, cold-clone route to proof authority, and public contribution protocol are described in sibling papers.

**Structure.** Section 2 gives the ordinary analytic criteria that reach supports of divergent reciprocal mass. Section 3 states and develops the finite-support denominator theorem, which settles no infinite support. Section 4 collects the further necessary conditions that rationality would force on an arbitrary infinite support. Section 5 places Theorem 1.2 beside representative supports already known to give irrational values, grouped by the argument that reaches them. Section 7 treats the squarefree support, whose values are known at every power-of-two base and which two of the arguments used here provably cannot reach at any even base. Section 8 gives the unrestricted and support-restricted topology and measure classifications, followed by the exact $1/2$ and $1/21$ membership criteria and unresolved branches. Section 9 states what remains.

## 2 Extensions beyond reciprocal summability

The following ordinary proofs exclude several classes with divergent reciprocal mass. Complete proofs, examples and countermodels appear in the analytic proof supplement. The variable-exponent argument below and the finite repair deadline are compositions developed from those arguments.

### 2.1 Divisibility-weighted and quantitative criteria

For a finite nonempty prime set $P$, let $h(a) = \prod_{p \in P} p^{v_p(a)}$. The weighted condition

$$\sum_{a \in A} \frac{h(a)}{a(2^{h(a)} - 1)} < \infty \tag{W}$$

is the hypothesis of the following criterion.

**Theorem 2.1** (divisibility-weighted supports)**.** *Let $P$ be a finite nonempty set of primes, let $A \subseteq \mathbb{N}_{>0}$ be infinite, and suppose (W) holds. Then $\sum_{a \in A}(b^a - 1)^{-1}$ is irrational for every integer $b \geq 2$, and the same conclusion holds for every infinite subset of $A$.*

The proof is ordinary and is given below and in the linked supplement; no formal check covers the assembled criterion. What the kernel covers is its analytic ingredients, listed at the end of this note. A fixed-base version replaces 2 by $b$ in the condition; arbitrary nested divisibility chains have the same criterion with $h(a)$ the largest chain element dividing $a$. The remaining obligation is unchanged: the criterion selects a class of reciprocal-divergent supports and leaves the rest of that regime open.

Here the decisive estimate is a finite orbit average. If $g = (a, Q)$, then

$$\frac{1}{T} \sum_{m=1}^{T} \frac{b^{Qm \bmod a} - 1}{b^a - 1} \leq \frac{g}{a(b^g - 1)} + \frac{1}{T(b^g - 1)}.$$

The second term cannot be discarded before summing over $a$. Choose $Q$ to freeze the finite prefix and all small $P$-parts. Average again over $T = 2^j$, $M \le j < 2M$. The incomplete-period contribution from small $P$-parts is at most $2QW/M$, where $W$ is the sum in (W); the large-part contribution is suppressed by the exponential denominator. This produces arbitrarily small positive shifted-atom displacements. The integral lattice gap then gives irrationality, and the binary atom comparison gives every larger base. The supplement gives all truncation estimates and an explicit reciprocal-divergent host with gaps $O(\log\log a)$.

The formal development also checks the summability of the dyadic observation mass, its vanishing mean, and the irrationality consumer from close returns. These are the checked analytic ingredients; the complete weighted criterion above still uses the ordinary assembly in the supplement.

There is also a quantitative necessary condition for rationality. Put $H_A(x) = \sum_{a\in A,\, a\le x} 1/a$, let $T_0 = 2$, $T_{j+1} = 2^{T_j}$, and set $\ell(x) = \min\{j : x \le T_j\}$. For a reduced rational value $p/q$ at base $b$, write $q = q_s q_c$, where every prime factor of $q_s$ divides $b$ and $(q_c, b) = 1$. Choose $0 \le c < q_s$ with $cq_c \equiv p \pmod{q_s}$, and put $\lambda = 1 - c/q_s$.

**Theorem 2.2** (sub-log-star reciprocal mass)**.** *Let $b \ge 2$ be an integer and let $A \subseteq \mathbb{N}_{>0}$ be infinite. If $\sum_{a\in A}(b^a - 1)^{-1}$ is the reduced rational $p/q$, then*

$$\liminf_{x\to\infty} \frac{H_A(x)}{\ell(x)} \ge \frac{\lambda}{2}, \tag{Q}$$

*with $\lambda$ as above. Consequently $H_A(x) = o(\ell(x))$ implies irrationality at every integer base, and the conclusion is hereditary under passage to infinite subsets.*

The proof is ordinary and is the displayed argument below; no formal check covers it. Indeed, at suitable multiples of the order of $b$ modulo $q_c$, rationality forces displacement at least $\lambda$. Applying the finite orbit estimate after freezing the prefix through $M$, and averaging $J$ dyadic lengths, gives, with $X = L2^{J-1}$,

$$J\lambda < (J + 2L)(H_A(X) - H_A(M)) + 4.$$

Taking $J$ a sufficiently large multiple of $L$ forces a mass increment arbitrarily close to $\lambda$. The LCM bound allows this at successive scales separated by two tower steps, giving (Q). Consequently $H_A(x) = o(\ell(x))$ implies irrationality at every integer base, even when the reciprocal sum diverges. The constant is a rational phase gap; no positive ordinary or logarithmic density is asserted.

## 2.2 Positive fractional divisor covers

Let $F_j$ be finite sets of positive integers and write $f_{F_j}(n) = \#\{a \in F_j : a \mid n\}$. Suppose that $0 < \alpha_j \le 1$ and $c_{j,d} \ge 0$ satisfy

$$f_{F_j}(n)^{\alpha_j} \le \sum_{d\mid n} c_{j,d}, \qquad C_j = \sum_{d\ge 1} \frac{c_{j,d}}{d}.$$

Set $B_j = 2^{\alpha_j}$.

**Theorem 2.3** (Variable-exponent positive covers)**.** *If*

$$\sum_{j\geq 1} C_j 2^{j\alpha_j} \frac{B_j}{(B_j-1)^2} < \infty,$$

*then every infinite subset of $\bigcup_j F_j$ has irrational Mersenne subseries at every integer base. The exponents may tend to zero.*

*Proof.* Let $U_F(n) = \sum_{r\geq 1} 2^{-r} f_F(n+r)$ and $V_j(n) = \sum_{r\geq 1} B_j^{-r} \sum_{d|n+r} c_{j,d}$. Subadditivity gives $U_{F_j}(n)^{\alpha_j} \leq V_j(n)$. For each positive integer $L$, exact divisibility frequencies give

$$\lim_{X\to\infty} \frac{1}{X} \sum_{m=1}^{X} V_j(Lm) = \sum_d \frac{c_{j,d}(d,L)}{d(B_j^{(d,L)}-1)} \leq \frac{C_j}{B_j-1}.$$

Fix $\varepsilon > 0$ and put $t_j = \varepsilon 2^{-j}$. The finite-average divisibility bound $(L+r)/d$ gives the summable dominator

$$\sum_{j>J} C_j t_j^{-\alpha_j} \left( \frac{L}{B_j-1} + \frac{B_j}{(B_j-1)^2} \right).$$

It justifies passing the average through all three sums. The limiting average of $\sum_{j>J} t_j^{-\alpha_j} V_j(Lm)$ is therefore at most $\sum_{j>J} C_j t_j^{-\alpha_j}/(B_j-1)$, which tends to zero as $J \to \infty$. Choose $J$ making it less than one, and then choose $L$ divisible by every member of $\bigcup_{j\leq J} F_j$. Arbitrarily large $m$ make the displayed sum less than one. At such points every remaining $U_{F_j}(Lm) < t_j$, so their total is less than $\varepsilon$. The frozen atoms have zero displacement. Infinitude gives strictly positive displacement, and the integral lattice argument excludes rationality. The binary comparison transfers the returns to all integer bases. □

The fixed-exponent criterion $\sum_j C_j < \infty$ follows by regrouping finite frames into blocks with exponentially decreasing costs. For a complete divisor frame $F = g\{d : d \mid M\}$, positive coefficients are provided multiplicatively by $\beta_\alpha(p^e) = (e+1)^\alpha - e^\alpha$. When $M$ is squarefree the optimal cost is

$$\frac{1}{g} \prod_{p|M} \left(1 + \frac{2^\alpha - 1}{p}\right).$$

This yields moving frames with divergent reciprocal mass. Choosing $\alpha_j = g_j^{-1/3}$ and prime blocks whose reciprocal product is about $\exp(g_j^{1/3})$ makes the variable criterion summable when $g_j \geq 16^j$; every fixed exponent fails for this particular frame decomposition. This does not exclude other fixed-exponent covers of the same union. Positivity matters: the divisor expansion for $F = \{2,3\}$ has coefficient $2^\alpha - 2 < 0$ at 6 when $\alpha < 1$.

**The prime-power regime.**

**Theorem 2.4** (prime-power supports)**.** *Let $A$ be an infinite subset of the prime powers, let $b \geq 2$ be an integer, and let $A'$ be a fixed dilation of $A$ or a finite modification of one. Then $\sum_{a\in A'} (b^a - 1)^{-1}$ is irrational.*

The argument is ordinary and carries no formal check. Its analytic input is Tao–Teräväinen [5, Theorem 3.1], so the analytic authority for this theorem is theirs and the thinning to every infinite subset is what is added here. The proof uses the reciprocal mass of the selected primes as its scale, so arbitrarily slow divergence is retained; higher prime powers produce errors only on prime-square events. The equidistribution, small-prime, progression and exceptional-set hypotheses are checked in the supplement. The full-prime theorem and the authors' stated full-prime-power extension are due to Tao and Teräväinen.

### 2.3 Fixed repair schedules, and what they decide

The repair mechanism requires neither periodic digits nor a rational target, and its uniform form is Theorem 1.5 in Section 1, stated there for every real $x \geq 0$ and kernel checked. The finite quantities below are computations against that criterion, and each one decides a schedule rather than a target.

Fixed multiplicative schedules have a different fate. Eighty exactly certified selected anchors through rank 727, together with Dirichlet's theorem, give infinitely many prime-cofactor increases of at least seven at multiplier 120 and at least two at multiplier 420. In particular, the proposed modulus-420 tetraprime repair inequality is false despite its earlier finite-window evidence. This leaves unrestricted cofinal repairs open. At $1/21$, combining the actually selected exponents 5 and 7 gives 60 favorable phase classes modulo 105 and strengthens the conditional upper-density bound for divergent quotient rows from $1/5$ to $2/7$. Using instead the twelve selected exponents $5, 7, 8, 9, 10, 18, 20, 24, 28, 42, 45, 60$ gives exactly 1011 favorable classes modulo 1260. The corresponding larger restricted divergence set has conditional upper density at least $337/840 > 2/5$. Its complete phase mask and an independent rational replay of the selected prefix are supplied with the synthesis. The older $2/7$ bound concerns the two-anchor restricted set. It proves neither target's membership. The complete phase certificate and the exact recurrence identities are in the synthesis supplement.

## 3 Finite-support denominator periods

A finite support has a rational value. The relevant arithmetic questions are the size of its denominator and the position of the base within that denominator.

For a finite nonempty $F \subseteq \mathbb{N}_{>0}$ and an integer $b \geq 2$, write

$$x_F(b) = \sum_{n \in F} \frac{1}{b^n - 1} = \frac{N_F}{D_F} \qquad (\gcd(N_F, D_F) = 1,\ D_F > 0).$$

We use the standard trivial-modulus convention $\operatorname{ord}_1(b) = 1$.

**Theorem 3.1** (finite-period noncollapse)**.** *Let $F \subseteq \mathbb{N}_{>0}$ be finite and nonempty, let $b \geq 2$ be an integer, and let $D_F > 0$ be the denominator of $x_F(b)$ in lowest terms. Then $D_F$ is coprime to $b$, and*

$$\operatorname{ord}_{D_F}(b) = \operatorname{lcm}\{n : n \in F\}.$$

*If moreover* $\operatorname{lcm}(F) \geq 2$, *then* $\operatorname{lcm}(F) < D_F$.

The order statement is checked, its reduced-denominator form is checked, coprimality is checked, and the growth clause is checked. Every clause is inside the kernel check. This theorem answers the second question exactly and, under its stated hypothesis $\mathrm{lcm}(F) \geq 2$, gives a strict lower bound for the first. The omitted boundary is genuine: at $b = 2$ and $F = \{1\}$ the two quantities are both 1. It sits here, and not among the leading results, because it settles no infinite support: it supplies no unbounded-support certificate, and no limit of it does.

Six instances at $b = 2$, which also show what the hypothesis $\mathrm{lcm}(F) \geq 2$ is for:

| $F$ | $x_F(2)$ | $D_F$ | $\mathrm{lcm}(F)$ | $\mathrm{ord}_{D_F}(2)$ |
|---|---|---|---|---|
| $\{1\}$ | 1 | 1 | 1 | 1 |
| $\{2\}$ | $1/3$ | 3 | 2 | 2 |
| $\{2,3\}$ | $10/21$ | 21 | 6 | 6 |
| $\{1,2,3\}$ | $31/21$ | 21 | 6 | 6 |
| $\{2,6\}$ | $22/63$ | 63 | 6 | 6 |
| $\{4,6\}$ | $26/315$ | 315 | 12 | 12 |

The last two columns agree in every row, which is the order statement. In the worked case $F = \{1,2,3\}$, the rational sum is $1 + 1/3 + 1/7 = 31/21$ and $2^6 \equiv 1 \pmod{21}$, whereas no smaller positive exponent gives 1. The first row is the only finite nonempty support with $\mathrm{lcm}(F) = 1$, and the only one on which $\mathrm{lcm}(F) < D_F$ fails; that is what the hypothesis $\mathrm{lcm}(F) \geq 2$ excludes.

The content is a noncollapse statement. Clearing denominators over $b^{\mathrm{lcm}(F)} - 1$ makes the period at most $\mathrm{lcm}(F)$ immediately; what is not immediate is that cancellation in the numerator cannot bring it below. The mechanism is that each selected exponent $n$ contributes, through cyclotomic factorisation, a prime-power modulus dividing $b^n - 1$ on which $b$ has order exactly $n$; no single reduction can remove all of them at once. Prime powers rather than primes is not a technicality. At $(b, n) = (2, 6)$ no prime divisor of $2^6 - 1 = 63$ has order 6 — one of the exceptional cases in Zsigmondy's theorem — while 9 does, and it is the prime power that carries the witness. The row $F = \{2,6\}$ above is that case in full: $D_F = 63 = 9 \cdot 7$, and since 2 has order 2 modulo 3 and order 3 modulo 7, the order 6 can only come from 9. The growth clause then follows formally, since the order divides $\varphi(D_F) < D_F$, where $\varphi$ is Euler's totient.

Two boundaries. This is an unconditional statement about every finite support, not a bounded table of examples, and not an implication with an open hypothesis; but it settles no infinite support, and no limit of it does. Denominator-period control is part of the classical method behind Erdős's 1948 argument [8]. Whether this exact sharp form appears in the literature has not been assessed, and no novelty is claimed.

## 4 Rational values and scaled tails

Section 5 below lists supports for which irrationality is known. The results here instead quantify over *every* infinite support whose value is rational, which is the quantifier in Problem 1.1. The base is 2 throughout, and each statement carries its own hypotheses.

**Integral scaled tails, and unboundedness.** Binary long division turns a hypothetical rational value into an integer recurrence. Fix an integer $v \geq 1$ and a sequence $f\colon \mathbb{N}_{>0} \to \mathbb{N}$ of coefficients. Call an integer sequence $u\colon \mathbb{N} \to \mathbb{Z}$ a *tempered scaled-tail sequence* for $f$ with multiplier $v$ if

$$u(n+1) = 2u(n) - vf(n+1) \quad \text{for every } n \geq 0, \qquad \text{and} \qquad u(n) = o(2^n).$$

The recurrence is one step of long division in base 2: double the remainder, then pay out the next coefficient. The growth condition is what pins the solution down, since two solutions of the recurrence differ by $C\,2^n$ for a constant $C$, and only $C = 0$ survives $u(n) = o(2^n)$. For any coefficient sequence with $f(n) \leq n$ — and $\mathrm{sc}_A(n) \leq d(n) \leq n$ qualifies, $d$ being the divisor function — the series $\sum f(n)2^{-n}$ is rational exactly when a tempered scaled-tail sequence exists for some $v \geq 1$, and every such sequence is then the scaled tail $u(n) = v\sum_{j\geq 1} f(n+j)2^{-j}$; hence exactly one such orbit exists (checked).

The next theorem exhibits such an orbit for $X_A$ itself, with the coefficient sequence shifted past the power of 2 in the denominator.

**Theorem 4.1** (unbounded scaled-tail states)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be infinite with $X_A = p/(2^c v)$, $v \geq 1$. Then there is $u\colon \mathbb{N} \to \mathbb{N}_{>0}$ with*

$$u(n) = v\sum_{j\geq 1} \mathrm{sc}_A(c+n+j)2^{-j}$$

*satisfying the exact recurrence*

$$u(n+1) + v\,\mathrm{sc}_A(c+n+1) = 2u(n), \qquad u(n) \equiv p\,2^n \pmod{v},$$

*and $u$ is unbounded.*

Checked as Lean. The useful clause is exact: these scaled-tail states cannot remain bounded. In particular they cannot eventually cycle through a finite set of states.

**Divisor coverage cannot have long gaps.** Say that $\mathrm{sc}_A$ has a *zero window* of length $h$ at $N$ if $\mathrm{sc}_A(N+1) = \cdots = \mathrm{sc}_A(N+h) = 0$, that is, if no element of $A$ divides any of $h$ consecutive integers. At $A = \{2,3\}$, for instance, $\mathrm{sc}_A$ vanishes exactly on the integers prime to 6, so every zero window has length at most 1: one of any two consecutive integers is even.

**Theorem 4.2** (sublogarithmic zero windows)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be nonempty with $X_A = p/(2^c v)$, $v \geq 1$. For every $\varepsilon > 0$ there is a constant $B$, depending on $A, p, \varepsilon, c$ and $v$, such that every zero window of $\mathrm{sc}_A$ at $c+N$ has length*

$$h \leq \varepsilon \log_2(N+1) + B.$$

*For fixed $A, p, c, v$ and $\varepsilon$, the same $B$ works uniformly in $N$ and $h$; no common constant over supports or numerators is asserted.*

Checked as Lean. Read as a constraint on counterexamples, this rules out zero windows of any fixed positive proportion of $\log_2 N$ once $N$ is large. The proof compares an exponential lower bound forced on the scaled tail with an upper bound for divisor sums that grows more slowly than any fixed power of $N$.

**A reciprocal-sum lower bound.** Let $\operatorname{ord}_v(2)$ denote the multiplicative order of 2 modulo an odd $v > 1$, and write the reciprocal mass of $A$ for $\sum_{a\in A} 1/a$.

**Theorem 4.3** (reciprocal mass bound)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be such that $\sum_{a\in A} 1/a$ converges, and suppose $X_A = p/(2^c v)$ with $v > 1$ odd and $\gcd(p, v) = 1$. Then*

$$\sum_{a\in A} \frac{1}{a} \geq \frac{1}{\operatorname{ord}_v(2)}.$$

Checked as Lean. A finite support already supplies an instance: $A = \{2, 3\}$ has $X_A = 10/21$, so $c = 0$, $v = 21$, $\gcd(p, v) = 1$ and $\operatorname{ord}_{21}(2) = 6$, and the bound reads $\frac{1}{2} + \frac{1}{3} \geq \frac{1}{6}$. The summability hypothesis is doing real work: a rational value with fixed odd denominator part $v$ and a convergent reciprocal sum requires mass at least $1/\operatorname{ord}_v(2)$. The statement does not apply when the reciprocal sum diverges, and it gives no positive lower bound uniform in $v$; it constrains the pair (support, denominator) jointly.

The critical dyadic case has a different endpoint.

**Theorem 4.4** (dyadic reciprocal-mass endpoint)**.** *Let $A \subseteq \mathbb{N}_{>0}$ be infinite and suppose $X_A = p/2^c$ for some $p \in \mathbb{Z}$ and $c \in \mathbb{N}$. Then the reciprocal support terms are not summable, or*

$$\sum_{a\in A} \frac{1}{a} > 1.$$

Checked as Lean. The two alternatives are an exact necessary consequence of a dyadic rational value. They do not contradict rationality by themselves: a contradiction requires separate proofs that the reciprocal terms are summable and that the mass is at most 1. The formal nonsummability alternative also does not, by itself, assert a particular asymptotic law for the partial sums.

**The Boolean–Möbius characterisation, stated earlier.** The theorems of this section give necessary conditions. The sharpest description of rational-valued supports the development has is an equivalence, and it is stated as Theorem 1.4 in Section 1, together with the admissibility conditions (2), the reconstruction $A = \{n : (\mu * Q_U)(n) = 1\}$ and the finite instance at $A = \{2, 3\}$. It belongs with this section's material and is placed among the leading results because it removes the support from the description altogether.

**Combined constraints on a rational counterexample.** Taken together: a counterexample to Problem 1.1 would have an unbounded scaled-tail sequence obeying an exact linear recurrence and sublogarithmic divisor-coverage gaps. By Theorem 1.2, its reciprocal mass must diverge. Theorems 4.3 and 4.4 remain useful conditional estimates for convergent reciprocal mass, but that regime can no longer contain a counterexample. Theorem 1.4 reconstructs every rational value, but by itself does not force the reconstructed support to be infinite. These qualifications matter: the statements have different hypotheses, and no jointly contradictory combination has been proved.

## 5 Representative known irrational supports

**Why periodic weights give divisibility.** The full-support and periodic-support results share one argument in the coefficients of (1). To see it, first take $m \geq 1$ and a purely $m$-periodic weight $w\colon \mathbb{N}_{>0} \to \mathbb{N}$ and put

$$c_w(n) = \sum_{d \mid n} w(d).$$

Fix a prime $p \nmid m$ and write a positive integer as $n = p^k u$ with $p \nmid u$. Grouping divisors by their $p$-adic valuation gives

$$c_w(n) = \sum_{d \mid u} \sum_{j=0}^{k} w(dp^j).$$

Euler's congruence $p^{\varphi(m)} \equiv 1 \pmod{m}$ makes each inner sequence periodic with a period dividing $\varphi(m)$. Choosing $k = b\varphi(m) - 1$ therefore gives $b$ copies of a complete cycle:

$$\sum_{j=0}^{b\varphi(m)-1} w(dp^j) = b \sum_{j=0}^{\varphi(m)-1} w(dp^j).$$

Thus $b \mid c_w(n)$. With $r$ distinct primes at these exact valuations, the divisor sum separates into $r$ such coordinates and gains a factor $b^r$. This is the arithmetic step that lets the certificate argument reach periodic supports: periodicity of the weights forces divisibility of their divisor transform. The full-support case is $w \equiv 1$.

Divisibility at one integer is not yet an irrationality proof. The checked argument arranges blocks of these valuations by the Chinese remainder theorem and controls the remaining weighted tail to obtain the required certificates. Its weighted form uses $w(d) = \sum_{i=0}^{W-1} \mathbf{1}_{\{w(d)>i\}}$ for a bound $W$ on $w$: each indicator layer is periodic, so the same divisor argument applies. The theorem cited in the table then clears a common denominator for rational weights and removes a finite prefix. Removing that prefix changes the series by a rational number; the periodic tail must still contain a positive weight. Without that hypothesis, a nonzero finitely supported sequence would give a rational counterexample.

The following table is representative, not exhaustive. It groups the displayed supports by five mechanisms rather than suggesting that its rows classify all known cases. In the sentence immediately following the p. 222 theorem, Erdős says that pairwise coprimality can be removed by a more complicated argument, but he does not give that argument; p. 226 repeats that boundary. The first row above is a local Lean theorem with its own LCM–Cesàro and integer-orbit proof. Novelty and priority are unassessed, and this checked proof is not identified with Erdős's omitted argument. The printed pairwise-coprime theorem remains represented as an attributed specialisation.

Under the additional hypothesis that the pairwise-coprime exponents are odd, Duverney, Suzuki and Tachiya strengthen this conclusion: Corollary 1 gives linear independence of (1) and three associated plus/minus reciprocal sums at every integer base

| Support $A$ | Bases | Mechanism and authority |
|---|---|---|
| *Reciprocal-summable supports* | | |
| arbitrary infinite $A$ with $\sum_{a\in A} a^{-1} < \infty$ | every $b \geq 2$ | Theorem 1.2; proved here |
| pairwise coprime, $\sum_{a\in A} a^{-1} < \infty$ | every $b \geq 2$ | Erdős [9], theorem on p. 222; retained as a separately formalised specialisation (formalised here) |
| *Classical full support, and base change into it* | | |
| all of $\mathbb{N}_{>0}$ | every $b \geq 2$ | Erdős 1948 [8]; formalised here |
| multiples of a fixed $d$ | every $b \geq 2$ | dilation: the multiples series at base $b$ *is* the full-support series at base $b^d$ (checked) |
| *Periodicity of the coefficient sequence* | | |
| infinite eventually periodic | every $b \geq 2$ | checked here; Luca–Tachiya prove the nonnegative purely-periodic case [6, Theorem 1, p. 139; proof pp. 149–150]; finite rational prefixes give the infinite eventual case |
| a residue class; the odd numbers | every $b \geq 2$ | special cases of the row above (residue class, odd); Luca–Tachiya's Example 2 strengthens the odd row to joint linear independence of every finite divisor-convolution ladder, also for negative integer bases with absolute value greater than one [6, Example 2, p. 140] |
| *Denominator gaps along a sparse support* | | |
| factorials $\{n!\}$ | every $b \geq 2$ | checked here |
| powers of two $\{2^k\}$ | every $b \geq 2$ | checked here |
| infinite pairwise coprime, $\sum_{a\in A} a^{-1} < \infty$ | every $b \geq 2$ | Erdős [9], theorem on p. 222; formalised here |
| *Multiplicative closure: Chowla–Erdős, refined* | | |
| $F_s(E)$ and the monomial images $\{jn^i : n \in F_s(E)\}$ | $b = q^j$ with $\lvert q\rvert^L \leq s$, $L = \mathrm{lcm}(1, \ldots, \ell)$ | Duverney–Tachiya [4, Cor. 1.2, author-preprint p. 4]; linear independence, not only irrationality; not formalised here |
| $s$-free positive integers, with or without 1 | $b = q^j$, $2 \leq \lvert q\rvert \leq s$, $j \geq 1$ | *ibid.*, with $E$ the primes and $\ell = i = 1$; deleting 1 changes the value by $1/(b-1) \in \mathbb{Q}$ |
| squarefree | every $b = 2^j$, $j \geq 1$ | Duverney–Tachiya [4, Cor. 1.2 and Ex. 1.1, author-preprint p. 4]; joint linear independence for every finite set of such bases; not formalised here |
| coprime to a fixed $N$; sums of two squares | every $b \geq 2$ | *ibid.*, Examples 1.3 and 1.2; not formalised here |
| *Analytic* | | |
| primes | $b = 2$ proved | Tao–Teräväinen [5], Thm. 1.3, p. 4; proof pp. 44–56; not formalised here |
| primes, $b \geq 3$; prime powers | — | *ibid.*, asserted as modifications with details omitted in the cited version |
| arbitrary infinite $A$ with $\sum_{a\in A} a^{-1} = \infty$ | — | **Open** (the remaining part of Problem 1.1) |

of absolute value greater than one [2, Corollary 1]. This is a stronger conclusion on a narrower support class; it does not remove the hypothesis from the row above.
Five remarks on the table: one on containment between the mechanisms, three on the literature rows, and one on what "checked here" means.

For the full-support row, the weighted certificate theorem is the exact checked statement; the support-series formulation displayed in the table is its later corollary.

*The mechanisms are not nested, and one non-containment is proved.* The base-$b$ full-support theorem is exactly the $A = \mathbb{N}_{>0}$ statement; the multiples rows genuinely specialise it after a base change, but the sparse rows do not. The distinction is intrinsic: the denominator-gap criterion behind the factorial and power-of-two rows *provably cannot* reach the full support: the prefix lcm grows too slowly for its hypothesis to hold (checked). Conversely the analytic method of [5] reaches the primes, which are neither eventually periodic nor pairwise coprime with summable reciprocals. No list contains another, and their union does not exhaust the infinite supports.

*One theorem of Duverney and Tachiya generates several rows.* Their refinement of the Chowla–Erdős method [4, Cor. 1.2, author-preprint p. 4; proof pp. 9–11] proves, for a pairwise coprime sequence $E$ of polynomial growth and the set $F_s(E)$ of products of its members with exponents below $s$, that 1 and the values $\sum_{n\in F_s}(q^{jn^i}-1)^{-1}$ are linearly independent over $\mathbb{Q}$ whenever $|q|^L \le s$, with $L = \operatorname{lcm}(1,\ldots,\ell)$ and $\ell$ bounding the exponent $i$. This is a row-generating theorem, not an isolated example.

With $E$ the primes, $s = \infty$ returns the full support at every base, while $s = 2$ returns the squarefree support. For $s = 2$, the constraint forces $\ell = 1$ and $|q| \le 2$, but the free exponent $j$ gives every base $b = 2^j$. Thus this theorem excludes bases that are not powers of 2, rather than all bases $b \ge 3$; it also excludes higher monomial degrees $i \ge 2$. Their conclusion is stronger than irrationality, being linear independence of the whole finite family across the chosen exponents $j$.

*The classical full-support value now has a digit-level refinement.* At base 2, Campbell writes the Erdős–Borwein constant in the equivalent forms

$$E = \sum_{n\ge1}\frac{1}{2^n-1} = \sum_{n\ge1}\frac{d(n)}{2^n}$$

and proves that the binary block 11 occurs infinitely often in its base-2 expansion [3, Theorem 1, p. 12; proof pp. 12–24]. This adds genuine digit-distribution information to the full-support row, but it neither proves normality nor addresses an arbitrary infinite support $A$, so it does not change the open status of Problem 1.1.

*The two analytic rows do not have the same standing.* Theorem 1.3 on p. 4, proved in Section 5 on pp. 44–56, of [5] proves the prime-support case at base 2, the series there being $\sum_{n\ge1}\omega(n)/2^n$. The extension to every integer base, and the prime-power support — which those authors themselves identify with Problem 1.1 — are asserted by remark, with the modifications explicitly left to the reader. The table keeps the three apart, and only the first is proved.

The full-support result formalises Erdős's theorem; Luca and Tachiya's RIMS paper already proves the nonnegative purely-periodic case; a finite rational-prefix correction gives the eventual-periodic extension used here [6, Theorem 1, p. 139; proof pp. 149–

150], while Theorem A restates the broader signed purely-periodic theorem without reproducing its earlier proof.

*Remark* (signed periodic weights: a gap in the method, not in the mathematics). Dropping nonnegativity weakens the conclusion available, and it is worth being exact about where. For $u\colon \mathbb{N}_{>0} \to \mathbb{Z}$ periodic and $b \geq 2$, what is checked here is a *dichotomy*: the series $\sum_{n\geq 1} u(n)/(b^n - 1)$ is irrational, *or* some power of $b$ times its value is an integer (checked). For a nonnegative support indicator the terminating alternative is excluded by the rows above; for mixed signs this argument does not exclude it.

The terminating alternative never in fact occurs. Theorem A in Luca and Tachiya's 2017 RIMS paper explicitly restates their earlier Theorem 1.1 in the exact form needed here: $\sum_{n\geq 1} a_n/(q^n - 1)$ is irrational for every integer $q$ with $|q| > 1$ and every nonzero purely periodic integer sequence $(a_n)$ [6, Theorem A, p. 139]. Thus the disjunction above is a consequence of the chosen argument, not a feature of the problem: the second branch is empty, and their author restatement is what shows it. The RIMS paper does not reproduce the earlier proof or verify the broader eventual-rational formulation, so neither is claimed here. The local statement retains only the interest of being an independently checked argument with a visible boundary. Section 7 records a second, sharper instance of the same phenomenon.

## 6 Structured supports: preserving the remaining divisibility

The periodic argument manufactures long divisible blocks. For a sparse composite support the difficulty is different: after finitely many support elements have been absorbed into a congruence modulus, what arithmetic freedom remains in the others? A finite-core bouquet makes that question exact. Write

$$A = E \cup \{c_i p_i : i \geq 0\}, \qquad c_i \mid Q, \quad e \mid Q \ (e \in E),$$

where $Q, c_i, e$ are positive, $E$ is finite, the $p_i > 1$ are pairwise coprime and coprime to $Q$, and $\sum_i p_i^{-1} < \infty$. The petals $p_i$ need not be prime. For every finite set $F$ of indices, put $M_F = Q \prod_{j\in F} p_j$. Then

$$\frac{c_i p_i}{\gcd(c_i p_i, M_F)} = p_i \qquad (i \notin F).$$

Thus a new ray retains its entire petal as the modulus governing its hits along a progression with step $M_F$ (reduced-modulus identity).

There are concrete infinite examples. Let $r_0, r_1, \ldots = 5, 7, 11, 13, \ldots$ be the primes after 3, take $Q = 6$, $E = \emptyset$, $p_i = r_i^2$, and alternate $c_i = 2, 3$. This gives $A = \{50, 147, 242, 507, \ldots\}$; every member has exactly three prime factors counted with multiplicity. The bouquet, infinitude and factor count are constructed, proved, and checked, respectively. The reciprocal-summable support theorem already settles these examples. The reduced-modulus identity describes their arithmetic structure; no additional block selector is needed for irrationality. The all-base argument is proved in the support-averaging supplement.

**Theorem 6.1** (unconditional orthogonal-petal criterion)**.** *Let $A$ be a bouquet as above. Then $\sum_{a\in A}(b^a - 1)^{-1}$ is irrational for every integer $b \geq 2$.*

*Proof.* Each positive core satisfies $c_i \geq 1$. The support decomposition gives

$$\sum_{a \in A} \frac{1}{a} \leq \sum_{a \in E} \frac{1}{a} + \sum_i \frac{1}{c_i p_i} \leq \sum_{a \in E} \frac{1}{a} + \sum_i \frac{1}{p_i} < \infty.$$

The exceptional sum is finite, and the last series is a defining bouquet hypothesis. The rays are distinct, so $A$ is infinite. Apply the all-base reciprocal-summable support theorem. In particular, the alternating-core example above gives an irrational sum at every integer base. □

The unconditional all-base bouquet theorem is now kernel checked. The older forced-slot selector is unnecessary for this corollary; it belongs to a stronger certificate construction.

**Theorem 6.2** (the composite-dilation defect)**.** *Let $A \subseteq \mathbb{N}$ and let $a, x \in \mathbb{N}_{>0}$ with $a \in A$. Define the foreign-divisor defect*

$$\delta_A(a, x) = \#\{d \mid ax : d \in A,\ d \nmid x,\ d \neq a\}.$$

*Then the divisor incidence satisfies the exact identity*

$$\mathrm{sc}_A(ax) = \mathrm{sc}_A(x) + \mathbf{1}_{a \nmid x} + \delta_A(a, x).$$

*If every member of $A$ is prime, then $\delta_A(a, x) = 0$, so the familiar prime-support dilation formula is recovered. More generally, if $A$ has an* `OrthogonalPetalBouquet` *witness $h_B$, then for every $i \in \mathbb{N}$,*

$$\delta_A(\mathrm{ray}_{h_B}(i), x) \leq |h_B.\mathrm{exceptional}| + \mathrm{sc}_{\mathrm{range}(h_B.\mathrm{petal})}(x),$$

*where* $\mathrm{ray}_{h_B}(i) = h_B.\mathrm{core}(i) h_B.\mathrm{petal}(i)$.

The first identity is the finite divisor partition checked at exact dilation identity. The hard point is that a composite multiplier can create support divisors which are neither the multiplier itself nor old divisors of $x$; they are recorded by $\delta_A$ rather than silently discarded. The defect vanishes under the prime-support hypothesis by prime-support no-defect, and the corresponding incidence formula is prime specialization. For a bouquet, every non-exceptional foreign divisor injects into a unique petal divisor of $x$, yielding the displayed budget through foreign-divisor classification and defect bound. The correction is concrete already for $A = \{2, 6\}$: multiplying $x = 1$ by the composite support element 6 creates the additional support divisor 2, so $\delta_A(6, 1) = 1$ (two-six witness).

Thus a composite bouquet cannot be analysed by copying the prime formula until this explicit foreign channel has been budgeted. This theorem provides that local budget. The unconditional irrationality in Theorem 6.1 follows instead from reciprocal summability, without estimating those orbit defects.

## 7 Squarefree support and a coordinate-dependent obstruction

Let $A_{\mathrm{sf}} = \{d \geq 2 : d \text{ squarefree}\}$: a support of density $6/\pi^2$, far denser than the primes, and not periodic. Its values at all power-of-two bases are jointly linearly independent with 1, by a theorem of Duverney and Tachiya recalled below. This section establishes that two block-certificate arguments used here cannot reach that support at any even base, and that the reason lies in a normalisation rather than in the value.

**Theorem 7.1** (squarefree divisor incidence)**.** *For every $n \geq 1$ the number of squarefree divisors of $n$ is $2^{\omega(n)}$, where $\omega(n)$ is the number of distinct prime factors, and hence*

$$\mathrm{sc}_{A_{\mathrm{sf}}}(n) = 2^{\omega(n)} - 1.$$

*In particular* $\mathrm{sc}_{A_{\mathrm{sf}}}(n)$ *is* odd *for every $n \geq 2$.*

The count is checked, the incidence formula is checked, and the parity conclusion is checked. The proof is the bijection between squarefree divisors of $n$ and subsets of its prime factors (Lean), so the count is $2^{\omega(n)}$; removing $d = 1$ leaves an odd number whenever $\omega(n) \geq 1$. At $n = 12 = 2^2 \cdot 3$ the squarefree divisors are $1, 2, 3, 6$, so $\mathrm{sc}_{A_{\mathrm{sf}}}(12) = 3$; at $n = 30$ they are the eight divisors of 30 and $\mathrm{sc}_{A_{\mathrm{sf}}}(30) = 7$.

### 7.1 The values are known at every power-of-two base

Duverney and Tachiya's Corollary 1.2, applied with $E$ the primes, $s = 2$ and $\ell = 1$, gives $L = \mathrm{lcm}(1) = 1$ and the admissibility condition $|q| \leq 2$. The exponent $j$ in their displayed family remains arbitrary. Thus, for every $h \geq 1$, their Example 1.1 says that the numbers

$$1, \qquad \sum_{n \geq 1} \frac{|\mu(n)|}{2^{jn} - 1} \quad (1 \leq j \leq h)$$

are linearly independent over $\mathbb{Q}$ [4, Cor. 1.2 and Ex. 1.1, author-preprint p. 4]. Since $|\mu|$ is the indicator of the squarefree integers including $n = 1$, and $A_{\mathrm{sf}}$ omits only $n = 1$, whose weight at base $2^j$ is $(2^j - 1)^{-1} \in \mathbb{Q}$,

$$X_{A_{\mathrm{sf}}}(2^j) = \sum_{n \geq 1} \frac{|\mu(n)|}{2^{jn} - 1} - \frac{1}{2^j - 1},$$

and rational translation preserves the joint linear independence with 1.

**Corollary 7.2** (joint power-of-two-base theorem)**.** *For every $h \geq 1$, the $h + 1$ numbers*

$$1, \quad X_{A_{\mathrm{sf}}}(2), \quad X_{A_{\mathrm{sf}}}(4), \quad \ldots, \quad X_{A_{\mathrm{sf}}}(2^h)$$

*are linearly independent over $\mathbb{Q}$. In particular, every $X_{A_{\mathrm{sf}}}(2^j)$, $j \geq 1$, is irrational.*

Two boundaries on that. It is a citation, not a formalisation: nothing in this development proves it. The admissibility condition $|q|^L \leq s$ at $s = 2$ allows only $|q| = 2$, so this citation covers the bases $2^j$ and does not cover bases that are not powers of 2; this note proves no result about those remaining values.

### 7.2 Two block-certificate hypotheses have no instance

Fix $b \geq 2$ and a nonnegative coefficient sequence $f$. The two hypotheses used below are best printed rather than named. For every precision $q \geq 1$, they ask for $N, K, L, C \in \mathbb{N}$ with $K \leq L$ satisfying the common conditions

$$\sum_{r=K+1}^{L} f(N+r)b^{L-r} \leq C, \qquad \exists t \in \mathbb{N} : f(N+L+1+t) > 0, \\ q(C+N+L+2) < b^L, \tag{3}$$

together with one of the two first-block conditions

$$\text{digitwise:} \quad b^r \mid f(N+r) \quad (1 \leq r \leq K), \\ \text{weighted:} \quad b^K \mid \sum_{r=1}^{K} f(N+r)b^{K-r}. \tag{4}$$

Thus the data $N, K, L, C$ may depend on $q$. The parity of Theorem 7.1 refutes both alternatives for the squarefree incidence sequence at every even base.

**Corollary 7.3** (neither divisibility-first hypothesis has an instance)**.** *For the squarefree support $A_{\mathrm{sf}}$ and every even base $b \geq 2$, neither the digitwise nor the carry-aware block-certificate hypothesis holds. Both fail already at precision $q = b^2$.*

*Proof.* The hypothesis quantifies over every precision, so it is enough to exhibit one precision at which no admissible data exist; we take $q = b^2$ and split on whether the first block is empty.

Both hypotheses ask, for every precision $q$, for $N$, $K \leq L$ and $C$ with a first-block condition on $\mathrm{sc}_{A_{\mathrm{sf}}}(N+1), \ldots, \mathrm{sc}_{A_{\mathrm{sf}}}(N+K)$, a middle bound $\sum_{r=K+1}^{L} \mathrm{sc}_{A_{\mathrm{sf}}}(N+r)b^{L-r} \leq C$, a nonzero coefficient beyond $N + L$, and $q(C + N + L + 2) < b^L$. Take $q = b^2$ and write $f = \mathrm{sc}_{A_{\mathrm{sf}}}$.

Suppose $K \geq 1$. The digitwise condition requires $b^r \mid f(N + r)$ for $1 \leq r \leq K$. If $N + K \geq 2$, its instance at $r = K$ makes the even number $b$ divide the odd number $f(N+K)$. The carry-aware condition requires $b^K \mid \sum_{r=1}^{K} f(N+r)b^{K-r}$; reducing modulo $b$ kills every term but $r = K$, so the same contradiction follows when $N + K \geq 2$.

The only remaining case with $K \geq 1$ is $N = 0, K = 1$. If $L = 1$, then $q(C+N+L+2) \geq 3b^2 > b$; if $L \geq 2$, the $r = 2$ term in the middle sum is $f(2)b^{L-2} = b^{L-2}$, so $qC \geq b^L$. Both contradict the strict size inequality.

Suppose $K = 0$, where both first-block conditions are vacuous. If $L = 0$ the middle sum is empty and $q(C + N + 2) < b^0 = 1$ is impossible. If $L \geq 1$ the $r = 1$ term gives $C \geq b^{L-1}$ whenever $N \geq 1$, and again the size inequality fails. If $N = 0$ and $L = 1$, its left side is at least $3b^2 > b$; if $N = 0$ and $L \geq 2$, the $r = 2$ term gives $C \geq b^{L-2}$ and hence $qC \geq b^L$. These exhaust the cases. □

The two formal nonexistence statements are checked directly as the carry-aware no-go and the digitwise no-go. The displayed proof is retained to expose the empty- and one-position-block boundary cases rather than leaving the scope hidden behind the interfaces.

### 7.3 The obstruction is a normalisation, not the value

Corollary 7.3 is a statement about two arguments failing on a value that Corollary 7.2 shows to be irrational. The question it raises is therefore not whether the value can be reached, but what the failure is a property *of*. It is a property of where the support starts.

Adjoin 1 to the support and write $A_{\mathrm{sf}}^{+} = A_{\mathrm{sf}} \cup \{1\}$, the full squarefree support. Then

$$X_{A_{\mathrm{sf}}^{+}}(b) - X_{A_{\mathrm{sf}}}(b) = \frac{1}{b-1} \in \mathbb{Q},$$

so the two supports pose the same irrationality question at every base, while the divisor incidence changes from $2^{\omega(n)} - 1$ to $2^{\omega(n)}$ — from odd to even at every $n \geq 2$. The parity obstruction of Corollary 7.3 evaporates under a shift that provably cannot change the answer. The shifted coefficient identity and the exact equivalence of the two irrationality questions are checked as $2^{\omega(n)}$ incidence and the shift iff.

More is true: the shifted first-block condition at base 2 asks for $2^{r} \mid 2^{\omega(N+r)}$, that is $\omega(N+r) \geq r$ for $1 \leq r \leq K$, and a Chinese-remainder construction reserving $r$ fresh primes for each shift $r$ supplies such an $N$ for every $K$. This is checked both as the arithmetic block theorem for $\omega$ and in the corresponding formal statement for the shifted incidence.

It does *not* follow that either argument certifies the shifted support: the opening block is one of the conditions listed above, and the middle bound and the arithmetic inequality are untouched by this observation. What does follow is a methodological point: an obstruction stated against a coefficient sequence can be an artefact of the normalisation chosen for that sequence. Before a no-go result is reported as a property of a problem, the coordinate it is stated in should be varied by a transformation the problem is known to be invariant under. Here the transformation is adding one rational number, and it removes the obstruction entirely.

The same invariance holds for every finite change, not only this one. If $A \triangle B$ is finite, choose $M$ above all of its elements. The two support series then have the same tail beyond $M$, while each omitted prefix is rational. The two directions of this argument are exactly the checked prefix lemmas Lean and Lean. Thus $X_A(b)$ is irrational if and only if $X_B(b)$ is irrational for every integer $b \geq 2$. The present shift is the smallest instance of a general checked finite-change principle.

This is the second time in this note that a boundary turns out to belong to the method rather than to the mathematics. In Remark 5 the terminating alternative of the signed periodic dichotomy is empty, and a theorem of Luca and Tachiya is what shows it; here a parity obstruction survives only until the support is shifted by one element. In each case the correction came from outside the development — once from the literature, once from asking what the statement was invariant under. A formalised no-go result carries exactly the authority of its hypotheses, and its hypotheses include the coordinates it was written in.

## 8 Achievement-set geometry and the value $1/2$

Problem 1.1 asks whether every value obtainable from infinitely many of the weights $w_n = (2^n - 1)^{-1}$ is irrational. The set of all values obtainable, from finite and infinite selections alike, is the achievement set $\mathcal{A} = \{\sum_{n\geq 1} \varepsilon_n w_n : \varepsilon_n \in \{0,1\}\}$ of Section 1, and this section is about its geometry.

**Theorem 8.1** (geometry of $\mathcal{A}$)**.** *$\mathcal{A}$ is compact, perfect, totally disconnected and nowhere dense, and has Lebesgue measure* 1*.*

Checked as compact, perfect, totally disconnected, nowhere dense, and of measure one. So $\mathcal{A}$ is a fat Cantor set: strict tail domination $\sum_{\ell>n} w_\ell < w_n$ opens a gap at every level, while the total measure is not lost. The classical antecedent is Kakeya's two-page paper: he proves perfectness for the subsum set of every absolutely convergent real or complex series and gives the real interval criterion; under the present strict tail inequality at every index, his result also gives nowhere density [1, p. 251]. The strict inequality, the resulting distinctness of subsums over distinct supports, and the Cantor conclusion for every integer base are also recorded in Remark 4.1 (p. 13) of Kovač and Tao [7]; no novelty is claimed for any of these facts. That remark makes no metric assertion.

Strict tail domination does not determine the measure: the weights $3^{-n}$ satisfy $\sum_{\ell>n} 3^{-\ell} = 3^{-n}/2 < 3^{-n}$ and produce a null achievement set, the base-3 digits-in-$\{0,1\}$ Cantor set. So the measure-one clause is added here rather than formalised from there, and it is the arithmetic of the Mersenne weights that supplies it.

With $T_n = \sum_{k>n} w_k$, the standard level-$n$ convex-hull cover consists of $2^n$ disjoint intervals of length $T_n$, its nested intersection is $\mathcal{A}$, and

$$2^n T_n = \sum_{j\geq 1} \frac{2^n}{2^{n+j}-1} \longrightarrow \sum_{j\geq 1} 2^{-j} = 1,$$

dominated by $2^{1-j}$. Continuity of measure from above therefore gives $\lambda(\mathcal{A}) = 1$.

Membership is characterised level by level, by a greedy expansion. Run through $n = 1, 2, \ldots$ carrying a remainder, initially the target $x$, and at level $n$ subtract $w_n$ from the remainder if $w_n$ does not exceed it, leaving the remainder unchanged otherwise; call a level at which nothing is subtracted *skipped*. Say that the expansion *survives* level $n$ if the remainder after that level is at most the remaining mass $T_n$. Then $x \in \mathcal{A}$ if and only if $x \geq 0$ and the expansion survives every level (Lean).

Non-membership of a nonnegative target is therefore visible at a single level, and for a rational target strict failure can be certified effectively: rational upper bounds for the remaining tail converge to $T_n$, so once one lies below the rational residual the fatal inequality is proved. The finite example below uses an exact rational upper bound. For $x = 3/4$, the greedy algorithm skips $w_1 = 1$ and leaves residual $3/4$, while the exact zero-lookahead upper bound for the remaining tail is $2w_2 = 2/3 < 3/4$; hence $3/4 \notin \mathcal{A}$ (Lean).

The selector-to-subsum map sending a $0/1$ string $(\varepsilon_n)$ to $\sum_n \varepsilon_n w_n$, which the strict tail inequality makes injective, remains injective after restriction to any subfamily. That matters because Problem 1.1 quantifies over every infinite support rather than over $\mathbb{N}_{>0}$. For a set $J$ of future offsets write $T_J(n) = \sum_{k\in J} w_{n+k+1}$ for the tail restricted to $J$ (definition, summable at Lean).

Then $T_J(n) < w_n$ for every $J$ and every $n \geq 1$ (checked), since deleting weights only shrinks a tail that already sits below $w_n$ at the full support. Consequently the digit map is injective on strings supported in $J$ (checked), stated on the subtype of digit strings vanishing off $J$ and evaluated by the restricted digit map, so the statement is about the subseries itself and not a projection of the full one. Thus uniqueness of the selector survives passage to an arbitrary subfamily; this is an injectivity statement, not a statement about binary digits of the value.

That is a constraint on the shape of an argument, not on the supports: it decides no value, and it is weaker than any statement in Section 4.

### 8.1 Geometry after restricting the allowed exponents

The injectivity statement has a geometric completion for every set $J \subseteq \mathbb{N}$ of allowed digit positions. Let

$$\mathcal{A}_J = \left\{ \sum_{k \in J} \varepsilon_k w_{k+1} : \varepsilon_k \in \{0, 1\} \right\}.$$

Formally, the allowed strings form the supported digit set, which is closed. Its range is the restricted achievement set; the range and image descriptions agree by the image theorem. Consequently $\mathcal{A}_J$ is compact and closed.

It lies inside $\mathcal{A}$ by support restriction and is therefore nowhere dense. If $J$ is infinite, the digit space is preperfect; injectivity transfers that property to $\mathcal{A}_J$, so the closed set is perfect. Thus every infinite allowed support gives a compact perfect nowhere-dense set with a unique selector for each point.

The metric classification is exact. The summability lemma controls digit terms, and changing one coordinate changes the value by precisely its signed weight by the update formula. If $k \notin J$, allowing $k$ splits the new set into $\mathcal{A}_J$ and its translate by $w_{k+1}$ exactly. The two pieces are disjoint, so adjoining one coordinate doubles the volume. At $J = \mathbb{N}$, the restricted set is the full set $\mathcal{A}$ by definition.

**Theorem 8.2** (volume dichotomy for every allowed support)**.** *For every $J \subseteq \mathbb{N}$, exactly one of the following measure formulas applies:*

$$\begin{aligned} &J = F^c \text{ for a finite } F, && \lambda(\mathcal{A}_J) = 2^{-|F|}, \\ &J^c \text{ is infinite}, && \lambda(\mathcal{A}_J) = 0. \end{aligned}$$

For finite $F$, the division-free identity multiplies the face volume by $2^{|F|}$ and its solved form gives $2^{-|F|}$. Monotonicity under enlarging $J$ is checked. It traps a set with infinitely many forbidden coordinates below finite-codimension faces of arbitrarily small dyadic measure, yielding measure zero. The two cases are assembled in the formal dichotomy. This theorem classifies the size of the value set generated inside any prescribed family of exponents. It does not classify the arithmetic nature of its individual points and therefore does not settle Problem 1.1.

### 8.2 The value $1/2$

A rational point of $\mathcal{A}$ attained by an infinite support refutes Problem 1.1. The distinguished candidate is $1/2$, and Theorem 8.3 gives a self-contained pair of exact alternatives.

The greedy expansion of $1/2$ is an exact finite computation as far as one cares to take it. Through level 21 it takes the exponents

$$2,\ 3,\ 6,\ 7,\ 14,\ 20,\ 21$$

and skips the others, the skipped runs being $\{1\}$, $\{4,5\}$, $\{8,\dots,13\}$ and $\{15,\dots,19\}$; every level through 21 is survived. The second condition below asks whether the list of skipped exponents is infinite, and no finite computation answers that.

In the statement, $u = (u_1,\dots,u_d) \in \{0,1\}^d$ is a finite prefix, $V(u) = \sum_{n\le d} u_n w_n$ is its value, and $T_n = \sum_{k>n} w_k$ as above.

**Theorem 8.3** (membership and non-membership at $1/2$)**.** *The value* $1/2$ *belongs to* $\mathcal{A}$ *if and only if its canonical greedy expansion omits infinitely many exponents. Dually,* $1/2 \notin \mathcal{A}$ *is equivalent to the existence of a finite* fatal gap *— a prefix* $u$ *through rank* $d$ *with* $V(u) + T_{d+1} < \frac{1}{2} < V(u) + w_{d+1}$*, which makes every continuation miss — and to the greedy orbit having a last skipped exponent. Moreover no finite support has value* $1/2$*.*

Checked as the greedy form, the terminal-bit form, the skipped-rank form, the fatal-gap equivalence, its transfer to non-membership, and the finite-support exclusion.

**Exact-row transport at the endpoint.** The formal argument exposes the mechanism behind one conditional half-membership implication. Write $r_{c-1}$ for the rational greedy remainder after ranks through $c-1$. Whenever $c \ge 4$ and

$$0 < r_{c-1} < w_c,$$

the skipped-core constructor first turns the finite greedy core into an exact local quotient row at endpoint $2c-2$, the local row constructor. The hard finite step is explicit: the deficit below $1/2$ is below the next Mersenne weight, so a Boolean word fills the entire upper-half capacity, the upper-half Boolean fill. The residual cannot vanish at any finite rank, the strict-positivity theorem; the odd denominator of every finite positive Mersenne sum rules out exact finite exhaustion of $1/2$. If positive skips occur cofinally in the exact sense the cofinal-skip hypothesis, their endpoints are cofinal by the row fan-in. The row-value estimate makes the discrepancy at endpoint $n$ at most $(n+1)/2^n$, and closedness then places $1/2$ in $\mathcal{A}$, the closed-set step.

No compatibility between the finite witnesses is assumed. The global forward endpoint theorem is in fact reversible.

**Theorem 8.4** (positive skips characterise the half-value)**.** *The condition* `CofinalPositiveHalfGreedySkips` *holds if and only if* $1/2 \in \mathcal{A}$*.*

This is the checked equivalence. Thus cofinal positive skips are not an independently weaker hypothesis: proving them already proves the half-membership endpoint, while half-membership forces them because finite residuals never vanish. Neither side is established here, so the theorem identifies the exact obstruction without supplying an unconditional counterexample to Problem #257.

**Theorem 8.5** (terminal scaled vanishing implies the half-value)**.** *Let $S$ be a* `HalfTerminalOnlyScaledVanishingSequence`*: its finite words exclude ranks* 0 *and* 1*, their depths tend to infinity, and the absolute terminal carry divided by* $2^M$ *tends to zero along the depths $M$. Then there is an infinite set $A \subseteq \mathbb{N}$ with*

$$\sum_{a \in A} \frac{1}{2^a - 1} = \frac{1}{2}.$$

*Consequently the universal irrationality assertion in Problem 1.1 is false under this hypothesis. This is a conditional implication, not a construction of $S$ and not a solution of Problem #257.*

The terminal-only argument first places $1/2$ in the achievement set by the vanishing normalized-carry estimate, then uses the finite-support exclusion to show that the limiting support is infinite: achievement-set conclusion, infinite-support lift. The problem-level refutation is the direct formal consequence rational half counterexample, universal-claim refutation.

The formal source also proves equivalent terminal-bit and unbounded skipped-rank formulations in its internal half-cylinder seam coordinate; the two middle source links above are those versions. Their definitions are not needed for the self-contained greedy and fatal-gap statement used here. The two sides are asymmetric: non-membership is witnessed by a *finite* fatal gap and is therefore semi-decidable, whereas membership is an infinite condition. Computing further can only raise a lower bound on where a fatal gap could occur; it cannot establish survival.

The next local obstruction is sharper than a numerical search. Suppose a take occurs at rank $b$, the next take at rank $c > b + 1$, and put $q = 2^b - 1$ and $m = 2^{c-1}$. If $R$ is the reciprocal of the residual just before the first take, then the final intervening skip is dyadically unsafe exactly when

$$\frac{q(m-1)}{q+m-1} < R < \frac{qm}{q+m}.$$

The interval has the exact width $q^2/((q+m)(q+m-1))$. For a single intervening skip, $c = b + 2$ and $m = 2q + 2$, giving the two-thirds band

$$\frac{q(2q+1)}{3q+1} < R < \frac{2q(q+1)}{3q+2},$$

whose width is below $1/9$. Thus the band is an exact localization of one greedy danger, not a claim that the actual orbit enters it.

**Theorem 8.6** (the sharp local two-adic obstruction)**.** *Let $p, D, q$ be positive odd integers. If*

$$q(2q+1)p < 2D(3q+1), \qquad 2D(3q+2) < 2pq(q+1),$$

*then $p \geq 7$. In particular, the odd numerator classes $p = 1, 3, 5$ cannot occupy the cleared two-thirds band. The bound is sharp for these local hypotheses: odd triples with $p = 7$ do occur in the band.*

The hard step is genuinely 2-adic. The band defect $\Delta = 6D - 2pq$ is positive and divisible by 4 when $p, D, q$ are odd, so $\Delta \geq 4$; combining that gap with the band inequalities forces $p > 6$. The exact formal boundaries are the general localization general band localization, the single-skip equivalence two-thirds band, and the sharp numerator conclusion odd numerator bound. Integral reciprocal states are excluded from the same band by integral safety. This is a one-step local no-go result: it neither proves that the half-greedy orbit reaches or avoids the band nor settles half-membership.

A different conditional argument would produce an infinite support of value $1/2$ from cofinal geometric stages.

**Theorem 8.7** (cofinal cylinders imply an infinite half-support)**.** *Suppose that for every $N$ there are $M, K$ with $\max\{N, 1\} \leq M$ and a nonempty* `CylinderStage` *$K$ $M$. Then there is an infinite set $A \subseteq \mathbb{N}$ with $0 \notin A$ and*

$$\sum_{a \in A} \frac{1}{2^a - 1} = \frac{1}{2}.$$

*Thus the universal irrationality assertion in Problem #257 is false under this cofinal-stage hypothesis. The formal proof does not construct the cofinal family of full-cylinder stages.*

The proof first converts each full-cylinder stage into a weaker terminal-only strip, then invokes closedness of the half-achievement set and the positive-support lift. The exact steps are terminal-only bridge, infinite half-support, and positive-support lift. This is a conditional implication, not evidence that such stages exist; proving or refuting their cofinality is the remaining task for this argument.

One natural construction for $1/2$ is ruled out. The Boolean support selected by the negative values of the Möbius function has value exactly $1/2$ plus the positive Möbius tail, hence at least $1/2 + 1/63$ (identity, bound). That closes the sign-truncation construction; it excludes no other support.

### 8.3 Five layers proved insufficient at $1/2$

The $1/2$ question has an exact reduction and an exact record of what cannot close it. With $\rho_s$ the seam integer greedy remainder, membership follows from $\rho_s \leq 2^s$ holding infinitely often (checked), and membership is equivalent to a cofinal late largest skip (checked).

*Remark* (what has been proved insufficient)*.* Five separate mechanisms are proved unable to close that reduction, each by a checked countermodel or bound: the row recursion is simultaneously silent at $s = 16$ while row 17 is good through skipped rank 14 (checked); the model $f(s) = 2^s + 2$ satisfies every row law and ceiling while never returning to the half point (checked); every ceiling growing at rate at least 4 is compatible with the descent law forever (checked); every ceiling propagating by one-step interval induction is $\Omega(4^s)$, so the target $\rho_s < 2^{s+1}$ is out of its reach (checked); and $2^s + 3$ is odd at every row, obeys the step law, and never reaches the half point, so no congruence modulo 2 empties a band of positive width (checked). A sixth observation removes the dyadic dichotomy: $2^{s+1} y_s = \rho_s + 2^s$ is an integer at every row (checked), so the orbit point is a dyadic rational at every finite row.

The measured trigger density of $\rho_s \le 2^s$ is 0.49967 over $s \in [5, 6000]$, with geometric gaps and maximum gap 13 at $s = 5372$. That figure is a computation over a bounded range and proves no cofinality. One correction is on record and is repeated here because it is the kind of statement a finite scan gets wrong: the residual invariant $\forall s\, d,\ \mathrm{seamResidualUpto}(s, d) < 2^{2s-d}$ was reported as empirically true with maximum ratio 0.99989 over $s \in [6, 3000]$, and it is false at $s = 13$, $d = 7$, where the residual is 524419 against the ceiling $2^{19} = 524288$ (checked). An independent rescan found that the sole failure in the stated range.

These are exclusions of named mechanisms. None of them decides whether $1/2 \in \mathcal{A}$, and the question stays open on both sides.

### 8.4 A second rational target

The target 1/21 has a useful property that does not depend on any finite search.

**Theorem 8.8** (finite-support obstruction at 1/21)**.** *For every finite set $F$ of exponents with $n \ge 2$ for all $n \in F$,*

$$\sum_{n \in F} \frac{1}{2^n - 1} \ne \frac{1}{21}.$$

This is checked by the formal theorem. The reduced denominator 21 has doubling order 6, so the exact denominator–lcm identity forces every selected exponent to divide 6. The lower bound $n \ge 2$ leaves only $2, 3, 6$, and the remaining eight subsets are discharged by exact rational arithmetic. Thus any representation of 1/21 along this rank-$\ge 2$ route, if one exists, must be infinite. The theorem proves nontermination only: it does not prove that 1/21 lies in $\mathcal{A}$.

A separate arithmetic lemma isolates the primitive cone that appears in one candidate expansion. Write

$$\mathcal{P}_{23}(n) = \{(p, q) \in \mathbb{N}^2_{>0} : \gcd(p, q) = 1,\ 2p + 3q = n\}.$$

Every $n \ge 11$ has a member of $\mathcal{P}_{23}(n)$ (checked), whereas $\mathcal{P}_{23}(10)$ is empty (checked). Multiplicity begins immediately at 11, and every $10k$ with $k \ge 2$ has at least two distinct primitive representations (rank eleven, multiples of ten). These are exact Diophantine facts, not a counterexample construction. In particular the recurring collisions show why primitive-cone coverage is not already a Boolean expansion: the missing step is a proof that the relevant integer multiplicities can be converted into zero–one reciprocal-Mersenne digits without changing the value.

### 8.5 The scaled-greedy criterion at 1/21

The finite-support obstruction makes 1/21 useful only if the infinite problem is stated in coordinates that retain the actual greedy orbit. Let

$$r_N = \mathrm{greedyMersenneRemainder}(1/21, N)$$

be the real remainder after rank $N$, put $y_N = 2^N r_N$, and set

$$c_{N+1} = \frac{2^{N+1}}{2^{N+1} - 1}.$$

The scaled recurrence is a nearly doubling map: it sends $y_N$ to $2y_N - c_{N+1}$ when $c_{N+1} \le 2y_N$, and to $2y_N$ otherwise. The latter is the exact lower branch, so rank $N+1$ is skipped precisely when $2y_N < c_{N+1}$.

Theorem 1.6 in Section 1 gives the trap equality, the exponential escape and the bounded cofinal return, for every nonnegative real target and with every clause kernel checked. At a rational target the lower branch supplies a further equivalent form, which is the one the $1/21$ discussion uses.

**Theorem 8.9** (rational lower-separatrix criterion)**.** *Let $q$ be a nonnegative rational and write $y_N(q) = 2^N \operatorname{greedyMersenneRemainder}(q, N)$. Then*

$$q \in \mathcal{A} \quad \Longleftrightarrow \quad \forall K\, \exists N \ge K : \qquad 2y_N(q) < \frac{2^{N+1}}{2^{N+1}-1}.$$

*In particular, $1/21 \in \mathcal{A}$ if and only if its scaled greedy orbit crosses this moving lower separatrix beyond every cutoff.*

The general rational criterion is kernel checked, and its $1/21$ specialization is kernel checked. The hard direction uses the rational normal form: membership is equivalent to infinitely many omitted exponents, and a skip is exactly a lower-separatrix crossing. This is an exact reformulation, not a proof that the crossings occur. It asks for neither convergence nor a prescribed uniform bound.

The quotient coordinate gives a complementary classification of what nonmembership would have to look like. Let

$$Q_N = \left\lfloor \frac{2^N}{21} \right\rfloor - P_N,$$

where $P_N$ is the integral numerator of the binary divisor-incidence prefix generated by that same greedy support. Thus $Q_N$ is the nonnegative denominator-21 defect after the six-periodic residue of $2^N \bmod 21$ has been removed. These are not independent models: the formal source proves an exact identity relating $Q_N$, $2^N r_N$ and a finite-prefix divisor tail (checked).

A finer denominator-specific coordinate works directly with the denominator-21 quotient-greedy orbit. At even depth $2R$, write

$$\begin{aligned} D_R &= \operatorname{twentyOneEvenQuotientGreedySupport}(R), \\ s_R &= \operatorname{twentyOneEvenQuotientGreedyRemainder}(R). \end{aligned}$$

Here $D_R \subseteq \{2, \dots, R\}$ is the Boolean support obtained by descending integer-greedy selection on the exact scaled quotient weights, and $s_R$ is its terminal scalar remainder (support, remainder). Write $\mathcal{F}_{21}$ for `TwentyOneFatalAlignedBranch`, the explicit branch in which the real greedy orbit has a fatal witness, only finitely many skipped exponents (hence cofinite eventual selection), eventual quotient/rational-greedy alignment, and eventual occupation of every doubling block.

**Theorem 8.10** (fatal-branch quotient refinement at $1/21$)**.** *The following statements hold.*

1. $1/21 \in \mathcal{A}$ *if and only if* $\mathcal{F}_{21}$ *does not hold.*

2. *If there is an unbounded sequence of ranks $R$ with $s_R \le 2^R$, then $1/21 \in \mathcal{A}$.*
3. *On $\mathcal{F}_{21}$, eventually $s_R > 2^R$, the boundary rank $R+1$ belongs to $D_{R+1}$, and $(D_R, s_R)$ follows one exact affine recurrence.*

The equivalence is kernel checked; the closed-row compactness step is kernel checked; and the eventual affine regime is kernel checked. Explicitly, on $\mathcal{F}_{21}$ the quotient orbit is eventually strictly supercapacity and loses its last Boolean choice: if $\tau_R$ is the periodic target pulse and $\pi_R$ the divisor pulse generated by $D_R$, then for every sufficiently large $R$,

$$s_R > 2^R, \qquad D_{R+1} = D_R \cup \{R+1\}, \qquad s_{R+1} = 4s_R + \tau_R - \pi_R - (2^{R+1} + 1).$$

The denominator-specific separation theorem additionally proves that every closed Boolean quotient row is exactly the canonical quotient-greedy row, including exact saturation at the boundary (kernel checked). An aligned crossing from saturation into strict supercapacity forces a missing canonical ancestor at one-third or two-thirds scale and a real skipped exponent with square-root-bounded defect (ancestor hole, scaled skip).

These conclusions remove alternative late Boolean branches; they do not exclude the full fatal/cofinite/aligned branch. On that branch the permanent affine-supercapacity recurrence is forced, so a contradiction of the recurrence would be sufficient, but the recurrence alone is not the exact membership endpoint. It is conditional, not a contradiction: synthetic pulses can sustain such an affine recurrence without being the actual divisor pulse.

# 9 Open problems

Problem 1.1 remains the frame. Theorem 1.2 settles every infinite support of convergent reciprocal mass, so the exact remaining universal boundary is the reciprocal-divergent case. The forced conditions in Section 4 are not known to be contradictory there. For the base-2 universal problem, three endpoint questions organise the remaining discussion. The universal problem and the half-value question are not independent, and the relation between them is asymmetric. No finite support has value $1/2$ (Theorem 8.3), so $1/2 \in \mathcal{A}$ would exhibit an *infinite* support with a rational value and thereby refute Problem 1.1; equivalently, a positive answer to Problem 1.1 puts $1/2$ outside $\mathcal{A}$. The converse direction does not follow: $1/2 \notin \mathcal{A}$ would give a finite fatal-gap witness and eliminate this candidate, but would not imply the universal statement. The half-value question is therefore a one-sided test of #257, not a second problem beside it.

**Problem 9.1** (membership of $1/21$ in the Mersenne achievement set)**.** Prove cofinal crossings of the exact moving lower separatrix

$$2\ \mathrm{scaledGreedyRemainder}(1/21, N) < \mathrm{mersenneScale}(N+1).$$

Equivalently, exclude the explicit fatal/cofinite/aligned branch $\mathcal{F}_{21}$. It is sufficient either to contradict the eventual permanent affine-supercapacity recurrence forced by that branch or to force an unbounded sequence of closed canonical quotient rows; neither sufficient route is claimed to be equivalent by itself.

1. **Problem 1.1 itself**, now only for infinite $A$ with $\sum_{a\in A} a^{-1} = \infty$. Section 4 constrains every hypothetical counterexample without excluding one. Every counterexample would have unbounded scaled-tail states, sublogarithmic divisor-coverage gaps and an admissible Boolean–Möbius scaled-tail sequence. No contradiction among the applicable constraints is known.
2. **Membership of $1/2$ in $\mathcal{A}$.** Theorem 8.3 makes this exactly equivalent to infinitely many greedy skips, and to a fatal gap on the other side; neither is proved. A proof of non-membership would take the form of one finite fatal gap.
3. **Problem 9.1.** Theorem 8.8 rules out finite support, while Theorem 8.10 reduces membership exactly to excluding $\mathcal{F}_{21}$. Contradicting its forced eventual affine-supercapacity recurrence or producing arbitrarily deep closed canonical rows would suffice. Neither input is known, and neither is promoted to an equivalence.

The half-value question remains valid on both sides; the criterion at $1/21$ is at present the sharper of the two. The questions below refine that criterion and the arithmetic constraints around it, in order of how directly an answer would move the proved boundary.

**Problem 9.2** (permanent affine supercapacity)**.** Can the *actual* denominator-21 greedy orbit satisfy $\mathcal{F}_{21}$ indefinitely? Equivalently, can the complete fatal, cofinite-selection, alignment and doubling-block hypotheses coexist with the eventual recurrence

$$s_R > 2^R, \quad D_{R+1} = D_R \cup \{R+1\}, \quad s_{R+1} = 4s_R + \tau_R - \pi_R - (2^{R+1} + 1),$$

or must an arbitrarily deep closed return $s_R \leq 2^R$ occur?

A contradiction from a Lyapunov function, a 2-adic obstruction, a divisibility theorem or a recurrence classification would prove $1/21 \in \mathcal{A}$ and, by Theorem 8.8, produce an infinite rational support. Conversely, an actual construction satisfying *all* clauses of $\mathcal{F}_{21}$ would prove $1/21 \notin \mathcal{A}$. Constructing only an abstract affine orbit with adversarial pulses does neither.

**Problem 9.3** (weakest native recurrence criterion)**.** Does the scaled actual greedy remainder return cofinally to one bounded interval?

$$\exists B < \infty \ \forall K \ \exists N \geq K : \qquad 2^N r_N \leq B.$$

This condition is both necessary and sufficient. It is the $x = 1/21$ specialization of the general membership equivalence in Theorem 1.6 (checked). It asks for neither convergence, a prescribed $B$, a global bound nor bounded return gaps. Two concrete stronger targets are cofinal recurrence of $Q_N \leq 1$ (sufficient condition) and arbitrarily deep closed quotient rows $s_R \leq 2^R$. Exact computation through rank 200,000 records 96 zero-defect returns and 4,956 returns to $Q_N \leq 1$, with last observed ranks 193,690 and 199,930 and maximum observed gap 492; these finite data do not prove cofinality.

**Problem 9.4** (actual-orbit invariant)**.** Is there a finite-memory, 2-adic or discrepancy invariant, using the correlated divisor pulses of the actual support, that forces a closed return or forbids permanent supercapacity? More precisely, can one use a bounded window of $R$ mod 6, residues of $s_R$, endpoint divisor counts and the finite set of eventual skips to force descent or a forbidden state? Alternatively, can one prove that no bounded-memory invariant distinguishes the true orbit from synthetic permanent-supercapacity controls?

Pointwise pulse bounds are known to be too weak. The exact six-step recurrence is

$$Q_{N+6} = 64Q_N + 3(2^N \bmod 21) - L_N,$$

with $L_N$ the actual weighted repair load (checked). The formerly plausible translation-invariant six-step contraction first fails at $N = 73$; the surviving statement is the slope-aware repair-load iff at the exact equivalence.

**Problem 9.5** (final-skip Diophantine exclusion). Let $E = \sum_{n\geq 1}(2^n - 1)^{-1}$. If $1/21 \notin \mathcal{A}$, let $M$ be the last skipped exponent, $S_M$ its finite skipped prefix, and

$$a_M = \frac{1}{21} + \sum_{d\in S_M} \frac{1}{2^d - 1}.$$

Can the complete final-skip signatures be used to prove $|E - a_M| \geq \mathrm{gap}_M$, contradicting

$$0 < a_M - E < \mathrm{gap}_M\,?$$

The fatal interval is checked as an exact one-sided approximation. The reduced skipped-prefix denominator has binary order equal to the lcm of the skipped exponents and hence at least $M$ (order identity, lower bound). Parity-sensitive gcd restrictions and adjacent-numerator product inequalities are also checked, but they are necessary signatures rather than the missing upper-height or irrationality-measure estimate.

**Problem 9.6** (classification of rational targets). For $L \geq 1$, classify the reduced rationals

$$\mathcal{T}_L = \left\{ \frac{p}{q} \in (0,1) : \begin{array}{l} \gcd(p,q) = 1,\ q \text{ odd},\ \mathrm{ord}_q(2) \leq L, \\ p/q \text{ has no finite Mersenne representation} \end{array} \right\}$$

for which non-membership reduces to an eventually deterministic finite-memory or affine quotient recurrence. In particular, classify $\mathcal{T}_6$ by target-pulse alphabet, finite-support obstruction, weakest membership criterion, surviving branch complexity and available Diophantine estimate. Is $1/21$ minimal or unique under explicit criteria, or does another target have a strictly simpler fatal branch?

**Problem 9.7** (realised finite denominators). For fixed $b \geq 2$ and $L \geq 2$, let $\mathcal{D}_b(L)$ be the reduced denominators of the nonempty finite sums $\sum_{n\in F}(b^n - 1)^{-1}$ with $\mathrm{lcm}(F) = L$. Is

$$\mathcal{D}_b(L) = \{D : D \mid b^L - 1,\ \mathrm{ord}_D(b) = L\}?$$

If not, what cyclotomic, valuation or cancellation conditions characterise the realised denominators?

The order theorem supplies the inclusion from left to right, not its converse. A counterexample and corrected classification, or effective extremal bounds within $\mathcal{D}_b(L)$, would strengthen the lead theorem and feed height information back into Problem 9.5. For scope, all squarefree values at power-of-two bases, jointly in each finite family, are settled by Corollary 1.2 and Example 1.1 of [4, author-preprint p. 4] (Corollary 7.2). Their cited corollary applies to power-of-two bases.

## Statements and declarations

The inline links identify the exact formal statements used in each section. The analytic criteria in Section 2 and the all-base bouquet composition in Theorem 6.1 have ordinary proofs above and in the linked supplement. The displayed proof of Corollary 7.3, the derivation of Corollary 7.2 from [4, Cor. 1.2 and Ex. 1.1, author-preprint p. 4], and the general finite-change argument (from the two checked prefix lemmas) are expository arguments rather than named checked statements.

The squarefree no-go interfaces, shifted coefficient, shift equivalence, Chinese-remainder block supply, and restricted-selector injectivity are directly checked statements linked at their use; they are not prose-only claims.

The numerical instances — the table of finite supports in Section 3, the incidence and zero-window examples, the reciprocal-mass and Boolean–Möbius instances, the 3/4 certificate, and the greedy expansion of 1/2 through level 21 — are direct computations, reported as computations and not as theorems. Results attributed to Campbell, to Duverney and Tachiya, to Tao and Teräväinen, to Luca and Tachiya, and to Kovač and Tao are cited from the literature and are not formalised here.

Erdős Problem #257 remains open.

**Checked declarations behind the reviewed support families.** Every finite-core orthogonal-petal bouquet has irrational reciprocal-power value at every integer base at least two. Its defining summable petal reciprocals bound the actual support reciprocals, support summability, so no forced-slot selector is needed: all-base bouquet corollary, specialising the base-two statement base-two bouquet theorem.

For the divisibility-weighted family the ordinary proof carries the full theorem; the kernel checks its analytic ingredients. The finite orbit estimate is Cesàro orbit estimate with the dyadic observation bound dyadic observation bound; for infinite supports the observation mass is bounded, mass bound, summable, mass summability, and has a mean limit, mean limit. The radix close-return endgame is close-return irrationality. The prime-part split and the diagonal scale producer remain ordinary steps.

For every real $x \geq 0$, with $Q_N$ the difference between $\lfloor 2^N x \rfloor$ and the Lambert prefix numerator of the actual greedy support, membership in the Mersenne achievement set is equivalent to a nonincrease of $Q$ in every window $[K, K + 2\lfloor \sqrt{K} \rfloor + 12)$, square-root window criterion, to cofinal nonincreases, cofinal repair criterion, and to cofinal bounds by the next divisor load, load bound criterion, or by $N + 1$, linear bound criterion. One strict-increase window excludes membership, strict-window exclusion; the target 4/9 is a specialisation.

## References


[1] S. Kakeya, *On the set of partial sums of an infinite series*, Proc. Tokyo Math.-Phys. Soc., 2nd ser., **7** (1914), 250–251. The perfectness theorem and the real tail criterion are on p. 251.

[2] D. Duverney, Y. Suzuki and Y. Tachiya, *Linear independence results for certain sums of reciprocals of Fibonacci and Lucas numbers*, arXiv:1908.07290, 2019. Corollary 1 gives

the stated sparse reciprocal-sum linear independence under the pairwise-coprime, odd-support and reciprocal-summability hypotheses.

[3] J. M. Campbell, *On the binary digits of the Erdős–Borwein constant*, arXiv:2605.24160v1, 2026. Theorem 1 on p. 12 proves that the binary block 11 occurs infinitely often in the base-2 expansion of the full-support value; its proof occupies pp. 12–24.

[4] D. Duverney and Y. Tachiya, *Refinement of the Chowla–Erdős method and linear independence of certain Lambert series*, Forum Math. 31 (2019), no. 6, 1557–1566, DOI. Corollary 1.2 gives the general $F_s(E)$ theorem and its monomial images under $|q|^{\mathrm{lcm}(1,\ldots,\ell)} \leq s$; Example 1.1 gives the joint squarefree family at all bases $2^j$, $j \geq 1$. Both are on p. 4 of the linked author preprint; the proof is on pp. 9–11.

[5] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, arXiv:2512.01739 (submitted December 2025, revised April 2026). Theorem 1.3 proves $\sum_{n\geq 1} \omega(n)/2^n = \sum_p (2^p - 1)^{-1}$ irrational, settling the prime-support case of #257 at base 2; the extension to every integer base and the prime-power support are stated as remarks with the modifications left to the reader. The theorem is on p. 4 and its proof is Section 5, pp. 44–56, in arXiv v2.

[6] F. Luca and Y. Tachiya, *Linear independence results for the values of divisor functions series*, RIMS Kôkyûroku No. 2014 (2017), 138–150. Theorem A on p. 139 explicitly restates their earlier Theorem 1.1 for nonzero purely periodic integer weights. Theorem 1 is on p. 139, its examples are on p. 140, and its proof is on pp. 149–150.

[7] V. Kovač and T. Tao, *On several irrationality problems for Ahmes series*, Acta Math. Hungar. 175 (2025), 572–608, DOI. Remark 4.1 (p. 13) records the strict tail inequality and the Cantor structure. Theorem 2.3 (p. 5; proof pp. 13–14) constructs rational merged sums from several bases under its mass hypothesis; it is not a fixed-base counterexample.

[8] P. Erdős, *On arithmetical properties of Lambert series*, J. Indian Math. Soc. 12 (1948), 63–66.

[9] P. Erdős, *On the irrationality of certain series*, Math. Student 36 (1968), 222–226 (issued 1969). The theorem on p. 222 treats pairwise-coprime support with convergent reciprocal sum at every integer base $b \geq 2$; the claimed removal of pairwise coprimality is stated without proof.

[10] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, Monogr. Enseign. Math. 28, Geneva, 1980, p. 62.

[11] P. Erdős, *On the irrationality of certain series: problems and results*, in A. Baker (ed.), *New Advances in Transcendence Theory*, Cambridge UP, 1988, pp. 102–109, doi:10.1017/CBO9780511897184.009.

[12] T. F. Bloom, *Erdős Problem #257*, `erdosproblems.com/257`, accessed 28 July 2026 (page displays "last edited 15 April 2026"). The current record labels the universal fixed-base problem open, cites `[Er68d]`, `[ErGr80, p. 62]` and `[Er88c, p. 105]`, records the Tao–Teräväinen prime and prime-power cases and the Kovač–Tao refutation of a broader varying-denominator speculation, and explicitly describes its status as the website owner's present assessment rather than a literature-completeness guarantee.